Supplementary Funding Request: ELN integration software for microscopy

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21 March, 2025

Executive Summary

In this proposal, the Data Science and Sharing Team (DSST) is requesting supplemental funds to expand on our work to make it easier for researchers to electronically track their experiments and analyses in a way that complies with the June 2024 mandate for all investigators to transition to electronic records and automate the packaging and publishing of experimental data to public data repositories. We are building a prototype of such a tool for Dr. Tejeda's section and have received positive feedback. Other investigators (e.g. Drs Alvarez, Langdon, and Penzo) have expressed their interest in similar functionality for their experimental workflow. Dr. Pereira and the Machine Learning Team (MLT) have also expressed an interest in standardizing the formats in which they receive data from investigators. The funds being requested here would allow us to direct external software developers to expand this tool to make it more robust and generalizable. [redacted]

Background & Prototype

As of June 30, 2024, all NIH IRP Investigators are required to use only electronic resources to document new and ongoing research, consistent with the NARA/OMB Federal wide Transition to Electronic Records mandate. Currently the most popular tool for complying with this mandate is LabArchives. While many investigators have found LabArchives to be a well designed tool, the graphical user interface can be somewhat clunky and labor intensive, especially when recording complex experimental pipelines that involve data acquisition, pre-processing, and analyses. The LabArchives Electronic Lab Notebook (ELN) comes with a basic API. The API can enable automatic reading and writing of data stored in notebooks. Using the API, a software developer can greatly increase the efficiency and transparency of both experiments and the analysis of the data and minimize the potential for human error. Dr. Josh Lawrimore of the DSST

has written a novel Python wrapper that facilitates the use of the Lab Archives API in Python. Currently, users granted access to a Lab Archives Electronic Notebook can use the wrapper to read and write notebook entries and add file attachments. Dr. Lawrimore intends to expand the Python wrapper to cover all API calls provided by Lab Archives. Here we use a set of experiments performed by Dr. Tejeda's section to demonstrate how a well-constructed application can reduce minutes of manual text entry into seconds of clicks in an application.

Dr. Hugo Tejeda's group conducts behavioral fiber photometry experiments on murine subjects. Using LabArchives' electronic lab notebook (ELN), fellows in Dr. Tejeda's section manually records experimental procedures using custom "Widgets" - HTML forms within the ELN web application. Dr. Lawrimore from the DSST has been working with Dr. Tejeda to build a prototype web application to make documenting and analyzing fiber photometry experimental pipelines semi-automated. Using this web application, a user can authenticate into Lab Archives and parse the entries for a given project in a notebook. The user can then create a series of folders on the local machine to house the experimental data described by the project entries. The structure of these folders will be dictated by the type of project, essentially creating a data standard so downstream analyses can be automated. Dr. Lawrimore confirmed that the data stored in the "Widgets" can be programmatically parsed and saved as an Excel spreadsheet, allowing automated transfer of information stored in an ELN to a local directory. The Python API wrapper and the current web application act as a prototype of the proposed tool below. Dr. Lawrimore has confirmed the basic functionality of the API wrapper to read and write data to and from an ELN.

Proposed Tool

The full functionality of the proposed application is as follows and in Figure 1. The proposed application will include a complete wrapper of the Lab Archive API to automatically parse Lab Archive notebook entries. Custom, modular interfaces will be capable of parsing information from Widget entries and simpler, standardized text-based entries. The information will then create a structured directory system on a specified data source (i.e. local data directory, OneDrive, Box, etc), with the organization dictated by the LabArchives entries. For example, when a subject's surgery is recorded, the application will generate a Surgery directory containing a copy of the completed Surgery Widget. This record enables the application to track which subjects have undergone surgery and are ready for behavioral experiments. Similarly,

records of behavioral experiments will create a Behavior directory with their corresponding completed Behavior Widgets. Thus, a link between the ELN entries and the storage area can be automatically established, a feature not previously available in the LabArchives Interface. The directory will be standardized and provide a valuable interface for facilitating automated analysis and tracking a study's progress.

For example, within the Behavior directory, users can store fiber photometry experiment results, including raw TDT tank files, AnyMaze protocol files, and videos, organized in cohort subdirectories. The proposed application will monitor these Behavior directories on the shared drive, allowing users to track which experiments have collected all necessary data and are ready for analysis. Beyond monitoring raw data, the application will initiate fiber photometry data extraction, pre-processing, and analysis pipelines. To ensure reproducibility, code versioning will be automatically recorded and documented in the LabArchives ELN.

The application will also automatically convert both metadata and data into the Neuroscience Without Borders (NWB) format, which is required for free data deposition in the Dandi Archive. When Dr. Tejeda's section is ready to share their data, the application will automatically upload the NWB-compliant data to Dandi Archive using their API. This use case demonstrates how the proposed application facilitates data sharing by leveraging ELN records and experimental data to automatically generate NWB-compliant datasets as experiments progress. The NWB datasets can be maintained in a completed state until publication. Since the application's goal is to package completed data automatically, researchers can selectively choose which data to include in the NWB format and add new data when needed, such as when addressing manuscript review comments.

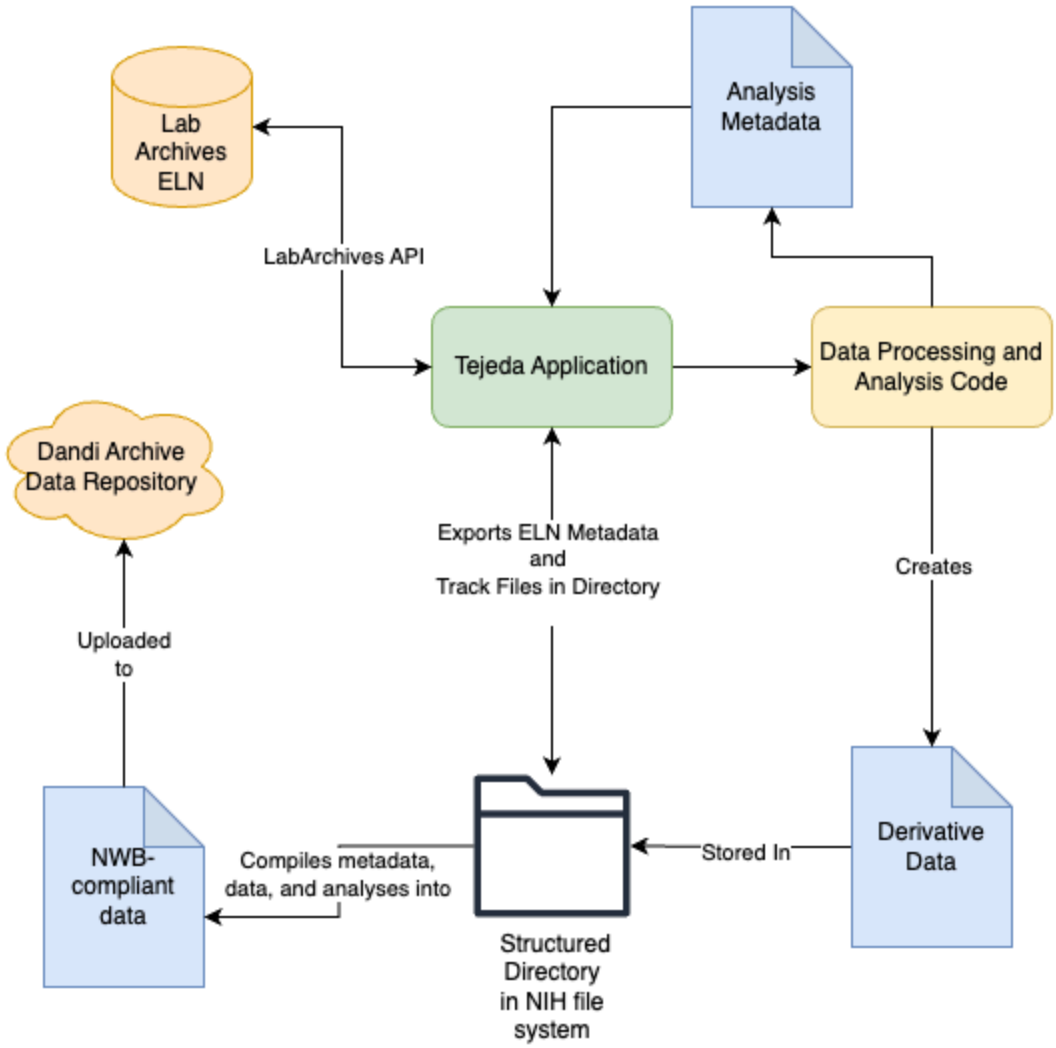


Figure 1. Flowchart of proposed application's connections for Tejeda use case. The application will sync information from Lab Archives Electronic Notebook (ELN) to the application for metadata parsing. The application will create a structured directory in a data location of the user's choosing. The application will sync the information between the structured directory and the ELN to keep project status and data packaging current. The application can also launch data preprocessing and analysis pipelines and record the data processing run in the ELN. Finally, the metadata, data, and analyses derivatives can be packed into an NWB data structure. The data can then be uploaded to the Dandi Archive data repository.

Third Party API Expansion and Modularization

In order to maximize the benefit of the proposed application, we plan on modularizing the proposed application (Figure 2). If an electronic notebook has an API, an interface can be written by the external software developers to allow the proposed application to connect to the notebook. Similarly, the software developers can create interfaces to allow the app-created directory to be stored in a variety of different data storage solutions (e.g., local desktop, remote server, cloud storage). The DSST and MLT, in conjunction with NIMH researchers, can direct the software engineers to support a variety of common experimental data formats, such as MRI, fiber photometry, and light sheet microscopy data. Lastly, the external software developers can write interfaces to allow the data to be packaged in either the Neurodata Without Borders (NWB) or the Brain Imaging Data Structure standards for automated upload to either NIH funded repositories such as the Dandi Archive or OpenNeuro.

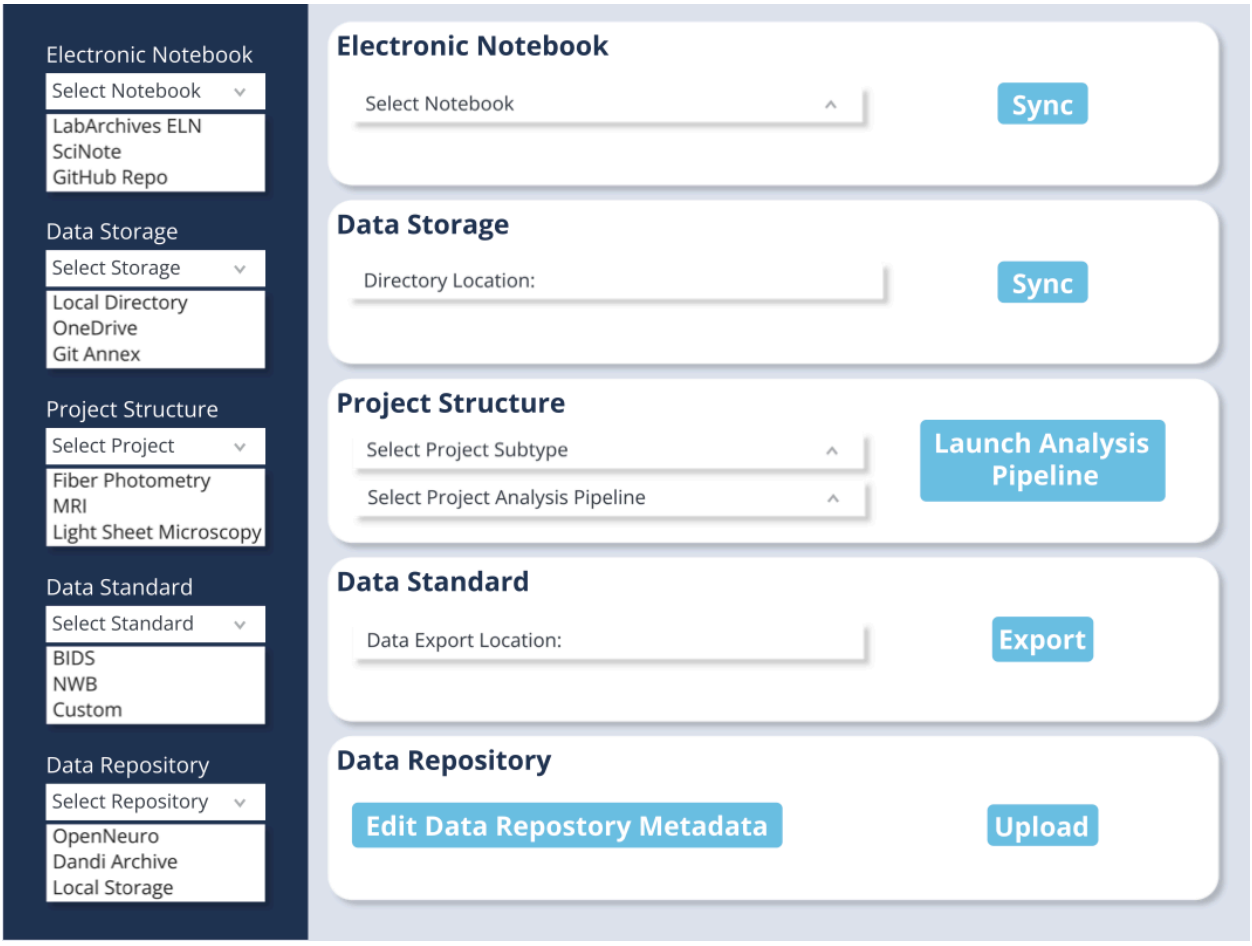


Figure 2. Mockup of proposed application. The main functions of the proposed application are represented in the mockup. A fully funded proposal would allow for the application to connect to various electronic notebooks platforms via their APIs, connect to a variety of data storage solutions for the creation and syncing of the project's structured directory, allow for a variety of project structures and analysis pipelines, package data into different data standards, and support uploading of the data to various data repositories. The modular nature of the application would allow for support of a variety of types of projects, not only the ones shown here.

